A Teaching Derivation of the Simple Harmonic Oscillator Propagator from the Lefschetz Thimble

Fixed endpoints $x_i = x_f = 0$, with the explicit example $m = 1$, $\omega = 2$, $T = 1$

1. Where we had got to

We had reached the point where the discrete oscillator matrix K has the same eigenvectors as the discrete Laplacian matrix L, and its eigenvalues are shifted by the potential term.

If

$$Lv^{(n)} = \mu_n v^{(n)},$$

then because

$$K = \frac{m}{a}L - am\omega^2 I,$$

we get

$$Kv^{(n)} = \lambda_n v^{(n)}, \qquad \text{with} \qquad \lambda_n = \frac{m}{a}\mu_n - am\omega^2.$$

Using the discrete sine-wave result

$$\mu_n = 2\left(1 - \cos\frac{n\pi}{N}\right) = 4\sin^2\left(\frac{n\pi}{2N}\right),$$

we obtain

$$\lambda_n = \frac{4m}{a}\sin^2\left(\frac{n\pi}{2N}\right) - am\omega^2.$$

These are the *mode stiffnesses*. So each discrete sine wave has its own quadratic coefficient.

This is the moment where the algebra starts becoming physically useful. Each eigenvector is one independent standing-wave shape for the path, and the corresponding eigenvalue tells you how costly that shape is in the action. A smooth low-frequency mode has a smaller stiffness, and a rapidly oscillating mode has a larger stiffness.

So now we stop thinking of the path as the list of time-slice positions

$$x_1, x_2, \ldots, x_{N-1},$$

and instead think of it as a superposition of standing waves. That is the natural language of the quadratic action.

2. Changing from position variables to mode variables

Let V be the orthogonal matrix whose columns are the normalised eigenvectors $v^{(n)}$. Then

$$V^T K V = \Lambda,$$

where

$$\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{N-1}).$$

Now define a new coordinate vector q by

$$x = Vq.$$

This is just a change of coordinates. The old coordinates x_j were the position values at each time slice. The new coordinates q_n are the amplitudes of the standing-wave modes.

Substituting this into the quadratic form gives

$$S_N[x] = \frac{1}{2} x^T K x = \frac{1}{2}(Vq)^T K (Vq) = \frac{1}{2} q^T (V^T K V) q = \frac{1}{2} q^T \Lambda q.$$

Since Λ is diagonal, this becomes

$$S_N[x] = \frac{1}{2} \sum_{n=1}^{N-1} \lambda_n q_n^2.$$

This is the key simplification. Originally the action mixed neighbouring time-slice variables together. After diagonalisation, the action is a sum of independent mode contributions.

So yes, this is exactly the step that removes the coupling. Not because the physics stopped coupling things, but because we moved into the coordinates in which the coupled system naturally decouples.

2.1. Why each eigenvector gives one independent quadratic mode

Suppose the path points entirely along one eigenvector direction, so that

$$x = qv.$$

If v is an eigenvector, then

$$Kv = \lambda v,$$

so the quadratic action becomes

$$S[x] = \frac{1}{2} x^T K x = \frac{1}{2}(qv)^T K (qv) = \frac{1}{2} q^2 v^T K v = \frac{1}{2} \lambda q^2 v^T v.$$

So along that direction the action is just a single square in the amplitude q. That is why each eigenvector gives one independent quadratic mode.

3. What the discretised propagator becomes after diagonalisation

The discretised propagator with fixed endpoints is

$$K_N(0, T; 0, 0) = \left(\frac{m}{2\pi i a}\right)^{N/2} \int_{\mathbb{R}^{N-1}} d^{N-1}x \, \exp\left(\frac{i}{2} x^T K x\right).$$

Now change variables from x to q. Because V is orthogonal, its Jacobian has absolute value 1, so the measure is unchanged,

$$d^{N-1}x = d^{N-1}q.$$

Therefore

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi ia}\right)^{N/2} \int_{\mathbb{R}^{N-1}} d^{N-1}q \, \exp\left(\frac{i}{2}\sum_{n=1}^{N-1} \lambda_n q_n^2\right).$$

Now the exponent is a sum of one-variable terms, so the integral factorises,

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi ia}\right)^{N/2} \prod_{n=1}^{N-1} \int_{-\infty}^{\infty} dq_n \, \exp\left(\frac{i}{2}\lambda_n q_n^2\right).$$

This is the precise point where the many-dimensional thimble problem turns into a collection of one-dimensional thimble problems.

4. The one-dimensional thimble

Now focus on one single mode,

$$I_n = \int_{-\infty}^{\infty} dq \, \exp\left(\frac{i}{2}\lambda_n q^2\right).$$

On the real line this is oscillatory. The phase keeps rotating as q changes, which is exactly the sign-problem picture, the arrows spin around.

For $\lambda_n > 0$, the steepest-descent contour is obtained by rotating by 45°,

$$q_n = e^{i\pi/4} y_n, \qquad y_n \in \mathbb{R}.$$

Why this angle? Because then

$$q_n^2 = e^{i\pi/2} y_n^2 = i y_n^2,$$

so

$$\frac{i}{2}\lambda_n q_n^2 = \frac{i}{2}\lambda_n (i y_n^2) = -\frac{1}{2}\lambda_n y_n^2.$$

So the oscillatory factor becomes

$$\exp\left(\frac{i}{2}\lambda_n q_n^2\right) \longrightarrow \exp\left(-\frac{1}{2}\lambda_n y_n^2\right),$$

which is now a real decaying Gaussian.

That is exactly what we wanted. The thimble has turned a wildly oscillating integral into an ordinary convergent Gaussian integral.

Also, the measure changes as

$$dq_n = e^{i\pi/4} dy_n.$$

So the one-mode integral becomes

$$I_n = e^{i\pi/4} \int_{-\infty}^{\infty} dy_n \, \exp\left(-\frac{1}{2}\lambda_n y_n^2\right).$$

Using the standard Gaussian formula

$$\int_{-\infty}^{\infty} e^{-ay^2/2}\,dy = \sqrt{\frac{2\pi}{a}}, \qquad a > 0,$$

we get

$$I_n = e^{i\pi/4}\sqrt{\frac{2\pi}{\lambda_n}}.$$

So each mode contributes a factor of this form.

5. Putting all modes together

Substituting back into the full product,

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi i a}\right)^{N/2} \prod_{n=1}^{N-1} \left(e^{i\pi/4}\sqrt{\frac{2\pi}{\lambda_n}}\right).$$

Collect the factors,

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi i a}\right)^{N/2} e^{i(N-1)\pi/4}(2\pi)^{(N-1)/2} \prod_{n=1}^{N-1} \lambda_n^{-1/2}.$$

But the product of the eigenvalues is the determinant,

$$\prod_{n=1}^{N-1} \lambda_n = \det K.$$

So

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi i a}\right)^{N/2} e^{i(N-1)\pi/4}(2\pi)^{(N-1)/2} \frac{1}{\sqrt{\det K}}.$$

This is a very important formula. It says that once you know the exact thimble of the quadratic problem, the whole path integral reduces to a determinant.

So the propagator is not being guessed, it is being computed exactly from the product of the mode contributions.

6. Why this gives the exact harmonic oscillator propagator

For the simple harmonic oscillator, the action is purely quadratic. That means the thimble method is exact, not approximate. There is no leftover interaction term to worry about. Every mode is Gaussian after the correct contour rotation, and the exact answer comes from multiplying all these Gaussian factors together.

When you evaluate the determinant, or equivalently take the continuum limit of the eigenvalue product, the result is the standard zero-to-zero harmonic oscillator propagator,

$$K(0,T;0,0) = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}}.$$

This is the same answer one gets from the Mehler kernel. So the thimble is not producing some alternative formula. It is providing the geometric explanation of why the oscillatory path integral collapses to this exact closed form.

6.1. Why there is no extra classical exponential here

In the general harmonic oscillator propagator there is usually an exponential factor involving the classical action,

$$K(x_f, T; x_i, 0) = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}} \exp\left[\frac{im\omega}{2\sin(\omega T)}\Big((x_f^2 + x_i^2)\cos(\omega T) - 2x_i x_f\Big)\right].$$

But here we are taking

$$x_i = 0, \qquad x_f = 0.$$

So that whole exponent becomes zero, and the exponential is simply 1. That is why, in this special case, the propagator is just

$$K(0, T; 0, 0) = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}}.$$

So for the zero-to-zero case, the entire answer is coming from the fluctuation determinant, which is exactly what the thimble calculation is evaluating.

7. Now put in the actual numbers

We now use

$$m = 1, \qquad \omega = 2, \qquad T = 1.$$

Then

$$\omega T = 2,$$

so

$$\sin(\omega T) = \sin 2.$$

Hence

$$K(0, 1; 0, 0) = \sqrt{\frac{1 \cdot 2}{2\pi i \sin 2}} = \sqrt{\frac{1}{\pi i \sin 2}}.$$

Now let us simplify this carefully. First write

$$\frac{1}{\sqrt{i}} = e^{-i\pi/4} = \frac{1-i}{\sqrt{2}}.$$

So

$$K(0, 1; 0, 0) = \frac{1}{\sqrt{\pi \sin 2}} \cdot \frac{1-i}{\sqrt{2}}.$$

Now evaluate the real numerical factor. Since

$$\sin 2 \approx 0.9092974268,$$

we get

$$\pi \sin 2 \approx 2.856642116.$$

Therefore

$$\frac{1}{\sqrt{\pi \sin 2}} \approx 0.591659983.$$

So

$$K(0,1;0,0) \approx 0.591659983 \cdot \frac{1-i}{\sqrt{2}}.$$

Divide by $\sqrt{2}$,

$$\frac{0.591659983}{\sqrt{2}} \approx 0.418366964.$$

Hence

$$\boxed{K(0,1;0,0) \approx 0.418366964 - 0.418366964\, i.}$$

That is the exact propagator value for the simple harmonic oscillator with these parameters.

8. What this means physically

This final complex number is the total sum over all paths from 0 to 0 in time 1. On the real contour, that sum is hard to see directly because different paths contribute complex phases that cancel each other. On the thimble, each normal mode has been rotated so that its contribution is a decaying Gaussian. So instead of adding huge numbers of spinning arrows that cancel, we integrate along the contour where each mode behaves cleanly and independently.

That is why the exact answer comes out so neatly. The thimble is the contour on which the oscillatory quantum problem becomes a real Gaussian mode-by-mode calculation.

9. The whole chain in one clean calculation

Starting from the diagonalised action

$$S_N[x] = \frac{1}{2}\sum_{n=1}^{N-1} \lambda_n q_n^2,$$

the propagator becomes

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi i a}\right)^{N/2} \prod_{n=1}^{N-1} \int_{-\infty}^{\infty} dq_n\, e^{i\lambda_n q_n^2/2}.$$

Rotate each mode to its thimble,

$$q_n = e^{i\pi/4} y_n,$$

so that

$$e^{i\lambda_n q_n^2/2} \to e^{-\lambda_n y_n^2/2}, \qquad dq_n = e^{i\pi/4} dy_n.$$

Thus

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi i a}\right)^{N/2} \prod_{n=1}^{N-1} \left[e^{i\pi/4}\sqrt{\frac{2\pi}{\lambda_n}}\right].$$

This is

$$K_N(0,T;0,0) = \left(\frac{m}{2\pi i a}\right)^{N/2} e^{i(N-1)\pi/4}(2\pi)^{(N-1)/2}\frac{1}{\sqrt{\det K}}.$$

Taking the exact harmonic oscillator continuum limit gives

$$K(0,T;0,0) = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}},$$

and then for

$$m = 1, \quad \omega = 2, \quad T = 1,$$

we get

$$\boxed{K(0,1;0,0) \approx 0.418366964 - 0.418366964\, i.}$$

10. Final takeaway

For the harmonic oscillator, the full contour problem is exactly solvable because the action is quadratic. So the true thimble is known analytically. That means that in the $\lambda = 0$ case, a machine-learned contour should recover this exact structure.

So when a numerical result at $\lambda = 0$ is close to

$$0.418366964 - 0.418366964\, i$$

for $m = 1$, $\omega = 2$, $T = 1$, start 0, end 0, that is not just rough agreement. It is the strongest sanity check that the contour-learning machinery is finding the right geometry.